GROUP PROJECT PROPOSAL (TEAM 9)
Fuel Price Analytics

Abstract

Fuel prices directly affect transportation costs, consumer budgets, and the overall economy. Our project will create a data analytics system that tracks fuel prices both historically and in real-time across different regions. We'll collect weekly data from the U.S. Energy Information Administration and live prices from gas stations via GasBuddy. Using Snowflake to store our data, Airflow to automate data collection, dbt to clean and organize the data, and Tableau to create dashboards, we'll build a complete pipeline from raw data to useful insights. We'll also include global price comparisons, oil market trends, and weather impacts to better understand what drives fuel price changes. Our final dashboards will help businesses and decision-makers understand pricing patterns, predict future prices, and spot market opportunities.

Dataset Links and Short Description

Primary Datasets:

1. U.S. Energy Information Administration (EIA) - Weekly Gasoline & Diesel Prices

Link: https://www.eia.gov/dnav/pet/pet_pri_gnd_dcus_nus_w.htm

Description: This dataset shows weekly U.S. gasoline and diesel prices going back to the 1990s. It's updated every Wednesday and comes directly from the government, so it's reliable. We'll use this as our main source of historical price data to find trends, seasonal patterns, and to build forecasting models.

2. Kaggle - Petrol/Gas Prices Worldwide Dataset

Link: https://www.kaggle.com/datasets/zusmani/petrolgas-prices-worldwide

Description: This dataset has fuel prices from countries around the world. We'll use it to compare U.S. prices with global prices, understand regional differences, and see how international markets affect domestic prices.

3. GasBuddy Real-Time Scraper API (via ScrapingBee)

Link: https://www.scrapingbee.com/scrapers/gasbuddy-scraper-api/

Description: This gives us live fuel prices from individual gas stations across the U.S., updated every hour. We'll use this to see price differences between regions and stations, spot unusual price changes, and create live dashboards that update in real-time.

Supporting Datasets:

1. Crude Oil Futures Data

Link: https://finance.yahoo.com/

Description: Oil futures prices are a leading indicator for gas prices—when oil prices go up, gas prices usually follow within 1-2 weeks. We'll track this to help predict future gas price movements.

2. Weather Data

Links: https://openweathermap.org/api and https://www.noaa.gov/

Description: We'll track weather events and extreme temperatures to understand their impact on fuel demand and supply chain disruptions (like hurricanes affecting refineries).

Reference Links

EIA Petroleum Data: https://www.eia.gov/dnav/pet/pet_pri_gnd_dcus_nus_w.htm

Kaggle Global Prices Dataset: https://www.kaggle.com/datasets/zusmani/petrolgas-prices-worldwide

GasBuddy Scraper API: https://www.scrapingbee.com/scrapers/gasbuddy-scraper-api/

Crude Oil Futures Data: https://finance.yahoo.com/

OpenWeather API: https://openweathermap.org/api

NOAA Weather Data: https://www.noaa.gov/

Snowflake Documentation: https://docs.snowflake.com/

Apache Airflow Documentation: https://airflow.apache.org/

dbt (data build tool) Documentation: https://docs.getdbt.com/

Tableau Public: https://public.tableau.com/

GitHub: https://github.com/